

OMNIKON NATIONAL HACKATHON 2026

PROJECT SUBMISSION

PhishGuard Bhasha

Regional-Language Phishing Detection

Problem ID	Omni_CyberTech_1
Domain	Cybersecurity, Blockchain & Digital Trust
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1. Problem Understanding

Phishing attacks increasingly target non-tech-savvy users through SMS, WhatsApp, and email messages written in regional Indian languages (Hindi, Bengali, Tamil, Telugu, Marathi, etc.), often mixed with English (“Hinglish” / code-switched text). Existing phishing detection tools are trained almost exclusively on English content, leaving a large and growing population underserved and vulnerable — particularly first-time internet users, elderly users, and residents of smaller towns who transact digitally but read primarily in their native language.

This gap has real financial and safety consequences: victims are tricked into sharing OTPs, bank details, or clicking malicious links disguised as government, banking, or delivery notifications in their own language, precisely because no existing tool warns them in a language and format they understand.

2. Proposed Solution

PhishGuard Bhasha is a web-based tool that detects phishing content — SMS text, WhatsApp-style messages, and URLs — written in regional Indian languages or code-mixed text, and warns users in real time before they click a malicious link or share sensitive information.

The system works in three steps:

- **Input:** User pastes a suspicious message or link into the app.
- **Analysis:** The system detects the language (including Hinglish/code-mixed text), classifies the text using a fine-tuned multilingual NLP model, and runs URL-based heuristic checks (suspicious domains, lookalike URLs, shortened links).
- **Output:** The user receives a clear risk verdict — Safe / Suspicious / Phishing — with a plain-language explanation in their own language (e.g. “This message asks for your OTP, which is a common scam pattern”).

3. System Architecture

The system follows a simple client-server architecture with a dedicated NLP inference layer:

- **Client Layer (Frontend):** A web app where users paste text/links and view results.
- **API Layer (Backend):** Receives input, orchestrates language detection, ML inference, and URL heuristic checks, and returns a structured risk response.
- **Language Detection Module:** Identifies the language of the input (Hindi, Hinglish, and additional regional languages).
- **Phishing Classification Model:** A fine-tuned multilingual transformer model that scores the text for phishing intent (e.g. urgency, credential requests, impersonation cues).
- **URL Heuristics Engine:** Checks extracted links against domain-reputation signals (lookalike domains, IP-based URLs, shortened links, blacklist APIs).
- **Response Layer:** Combines model output and heuristic checks into a single risk score and human-readable explanation returned to the user.

Data flow: User Input → Language Detection → [Text Classifier + URL Heuristics run in parallel] → Risk Aggregation → Explained Verdict to User

4. Key Features

- Multilingual phishing detection covering Hindi and Hinglish (code-mixed) text at MVP, with additional regional languages as stretch goals.
- Instant risk verdict (Safe / Suspicious / Phishing) with a confidence score, returned in under 2 seconds.
- Plain-language explanations in the user's own language, describing why a message was flagged (e.g. OTP requests, urgency tactics, impersonation of banks/government bodies).
- URL safety checks for lookalike domains, IP-based links, and known malicious domains alongside text analysis.
- Accessible, low-literacy-friendly UI designed for first-time and elderly users, with minimal jargon and clear visual cues.
- Privacy-conscious processing — message content is analyzed and not stored beyond the session unless the user opts to contribute it as a training sample.

5. Technology Stack

Layer	Technology
Frontend	React + Tailwind CSS
Backend / API	Python (FastAPI)
Language Detection	fastText / langdetect language-ID model
Phishing Text Classification	Fine-tuned multilingual transformer (MuRIL / IndicBERT / XLM-RoBERTa)
URL Heuristics	Domain-reputation checks, Levenshtein-distance brand matching, Google Safe Browsing / PhishTank APIs
Hosting / Deployment	Vercel or Render (frontend + backend), lightweight model inference endpoint
Data	Augmented/translated public phishing & SMS-spam datasets + curated Hindi/Hinglish samples

6. Implementation Plan

Phase	Timeline	Deliverable
Research & Data Collection	Aug 19–21	Dataset assembled (translated + hand-curated Hindi/Hinglish samples), tech stack finalized
Model & Backend Development	Aug 22–25	Trained multilingual classifier, working API endpoint, URL heuristics module

Phase	Timeline	Deliverable
Frontend & Integration	Aug 26–29	End-to-end working web app connecting UI to backend, UI polish for accessibility
Testing & Refinement	Aug 30–Sep 1	Bug fixes, accuracy testing on curated demo dataset, stretch-goal features, final pitch materials, submission by Sep 1

Key Risks & Mitigations

- **Limited labeled regional-language phishing data** will be addressed through translation and augmentation of existing English datasets, combined with manually curated samples.
- **Overfitting risk on a small dataset** is mitigated by fine-tuning an existing pretrained multilingual base (e.g. MuRIL/IndicBERT) rather than training from scratch.
- **Scope discipline:** scope is intentionally limited to Hindi and Hinglish at MVP to ensure a working, reliable demo within the compressed timeline.
- **Compressed schedule:** with submission due September 1, phases run back-to-back with no buffer days; the frontend and backend tracks overlap where possible to save time.